

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Any two (×1) from: Flame {mouth of tubes / pipette} (1) Hold lid above base of petri dish/ lift lid {slightly / at an angle} (1) Use a sterile pipette/ pipette tip (1) Work close to (the updraft of) a Bunsen burner (1)	2			2		2
		(ii)		$6.4 \times 10^5 = 2$ marks If incorrect award one mark for $2 \times 32 \times 10000$ 64×10^4 3.2×10^5 (not multiplied by 2) 640000		2		2	2	
	(b)	(i)		$5 = 3$ marks If incorrect award 2 marks for 4.9833887043 (incorrect rounding / not to whole number) If incorrect award 1 mark for $N = \frac{7-4}{0.301 \times 2} \quad (1)$ $N = \frac{3}{0.602}$		3		3	3	
		(ii)		(to give time to) synthesise {enzymes / proteins / DNA} / activating genes Ignore acclimatising to conditions	1			1		
		(iii)		(OD method) counts dead cells as well as living / total count for OD and viable count for the plates (1) For first 40 minutes {no / few} dead cells / owtte (1) As time passes there are more dead cells (1)			3	3		
				Question 5 total	3	5	3	11	5	2